

Sub-Models & Recursive Manifests: Post-Fix Assessment

Generated 2026-07-18 after fixing `derive_interfaces()` and capability filtering.

1. What Was Fixed

Fix 1: `derive_interfaces()` now uses `imports_detailed`

Before: `derive_interfaces()` only matched against `mod.imports` (bare names like `pathlib`, `types`). These never matched project modules because `module_to_file` had dotted paths like `src.architecture_model.core.types`. Result: **0 interfaces** across all manifests and sub-models.

After: Added a second pass that iterates `imports_detailed`, resolving:

- Relative imports: `from .types import X` → same-dir resolution, `from ..core.parser import X` → parent-dir resolution
- Absolute imports: `architecture_model.core.parser` → tries with `src/` prefix
- Package imports: `architecture_model.core` → resolves to `core/__init__.py`

Result: **83 interface edges** from full project scan.

Fix 2: Capability filtering

Before: Every function became a Capability, including `_parse_raw`, `_build_id_map`, etc. F5 (Manifest) had 16 capabilities.

After: Two-stage filter:

1. Skip functions starting with `_` (private helpers)
2. If block's `__init__.py` has exports, only promote exported functions
3. Fall back to all non-underscore functions if no exports found

Result: F5 went from 16 to **8 capabilities** (the actual public API).

2. Sub-Models: Before vs After

Block	Caps Before	Caps After	Ifaces Before	Ifaces After
F1 (Cli)	1	1	0	0
F2 (Config)	4	3	0	3
F3 (Core)	20	20	0	7
F4 (Extract)	1	1	0	0
F5 (Manifest)	16	8	0	22
F6 (Profiles)	1	1	0	0
F7 (Spec)	0	0	0	0
F8 (Utils)	4	4	0	0

3. Detailed Sub-Model Contents

F1: Cli

- **Refines:** COMP-CLI
- **Components:** 1
- **Capabilities (1):** main
- **Interfaces:** 0
- **Relationships:** 0

F2: Config

- **Refines:** COMP-CONFIG
- **Components:** 1
- **Capabilities (3):** load_config | discover_config | get_config
- **Interfaces (3):**
 - init -> schema
 - init -> loader
 - loader -> schema
- **Relationships:** 0

F3: Core

- **Refines:** COMP-CORE
- **Components:** 7
- **Capabilities (20):** coverage_report | compute_complexity | identify_systems | auto_assign_f_blocks | decompose_model | test_affinity_decompose | diff_models | merge_manifest | enrich_from_manifest | compact_for_generation | compose_enriched_model | load_model | validate_model_data | dump_model | save_model | slice_by_fblock | slice_by_layer | slice_by_status | slice_for_artifact | validate_model
- **Interfaces (7):**
 - coverage -> types
 - decomposer -> types
 - differ -> types
 - merger -> types
 - parser -> types
 - slicer -> types
 - validator -> types
- **Relationships:** 13

F4: Extract

- **Refines:** COMP-EXTRACT
- **Components:** 1
- **Capabilities (1):** extract_from_code
- **Interfaces:** 0
- **Relationships:** 0

F5: Manifest

- **Refines:** COMP-MANIFEST
- **Components:** 8
- **Capabilities (8):** process_block | print_summary | generate_manifest | load_or_generate_manifest | derive_interfaces | compute_metrics | scan_file | get_manifest_slice
- **Interfaces (22):**
 - **init** -> blocks
 - **init** -> display
 - **init** -> generator
 - **init** -> interfaces
 - **init** -> metrics
 - **init** -> scanner
 - **init** -> slicers
 - **init** -> types
 - ... and 14 more
- **Relationships:** 18

F6: Profiles

- **Refines:** COMP-PROFILES
- **Components:** 1
- **Capabilities (1):** load_profile
- **Interfaces:** 0
- **Relationships:** 0

F7: Spec

- **Refines:** COMP-SPEC
- **Components:** 1
- **Capabilities:** 0
- **Interfaces:** 0
- **Relationships:** 0

F8: Utils

- **Refines:** COMP-UTILS
- **Components:** 1
- **Capabilities (4):** is_excluded_dir | collect_py_files | discover_source_files | discover_test_files
- **Interfaces:** 0
- **Relationships:** 0

4. Cross-Block Dependency Graph

Block	Depends On
F1 (Cli)	F2, F3, F5
F2 (Config)	F8
F3 (Core)	F2, F5, F6, F8
F4 (Extract)	F2, F3, F5
F5 (Manifest)	F2, F3, F8
F6 (Profiles)	(none)
F7 (Spec)	(none)
F8 (Utils)	(none)

5. Remaining Gaps

- **F1 (Cli) has 0 interfaces.** The CLI uses lazy imports inside function bodies (from `..core.parser import load_model` inside `_cmd_validate()`). These are not captured by the AST scanner as module-level imports. Low priority – the CLI is a thin dispatch layer.
- **F3 (Core) capabilities not filtered by exports.** Core's `__init__.py` is empty, so the fallback includes all 20 non-underscore functions. Would need `__all__` added to `core/init.py` to filter further.
- **F7 (Spec) has 0 everything** except 1 component. It contains only `schema.json` (no Python functions). This is correct.